HW2.1

SCIE 001 Math

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1. Suppose that f is differentiable with $f'(x) > 0$ for all x and suppose that $f(1) = 0$. Set

$$F(x) = \int_0^x f(t)\,\mathrm{d}t$$

Determine if each statement is True or False. Justify your answer.

(a) The function $F(x)$ is continuous.

True

Taking the derivative of both sides:

$$\frac{\mathrm{d}}{\mathrm{d}x}F(x) = \frac{\mathrm{d}}{\mathrm{d}x}\int_0^x f(t)\,\mathrm{d}t$$

By the Fundamental Theorem of Calculus:

$$\frac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$$

We know $f(x)$ exists for all x, hence $\frac{\mathrm{d}}{\mathrm{d}x}F(x)$ must exist for all x, therefore $F(x)$ is differentiable for all x. Since differentiability implies continuity $F(x)$ must therefore be continuous for all x.

Therefore $F(x)$ is continuous for all x.

A different proof is given below for fun.

For $F(x)$ to be continuous for all x:

$$\lim_{a\to x} F(a) = F(x)$$

Therefore we must prove:

$$\lim_{a\to x}\int_0^a f(t)\,\mathrm{d}t = \int_0^x f(t)\,\mathrm{d}t$$

To do this, we can show that we can rearrange the left side to equal the right side.

Expanding the left integral to a Riemann Sum gives:

$$\begin{aligned}
&\lim_{a\to x}\left(\lim_{n\to\infty}\sum_{i=1}^{n}\frac{a-0}{n}f\left(\frac{i(a-0)}{n}+0\right)\right)\\
&= \lim_{a\to x}\left(\lim_{n\to\infty}\sum_{i=1}^{n}\frac{a}{n}f\left(i\frac{a}{n}\right)\right)
\end{aligned}$$

Since we know $f(x)$ is differentiable, it must also be continuous, as differentiability implies continuity. For any $n \in \mathbb{R}, n \neq 0$, $\frac{a}{n}$ is continuous for all a. Consequently, for any $i \in \mathbb{R}$:

$$\frac{a}{n}f\left(i\frac{a}{n}\right)$$

is also continuous for all a.

The sum of any continuous functions is also continuous. Therefore:

$$\sum_{i=1}^{n} \frac{a}{n} f\left(i \frac{a}{n}\right)$$

is continuous for all a. More precisely, this expression is continuous for all a independently of the value of n. Therefore,

$$\lim_{n \to \infty} \sum_{i=1}^{n} \frac{a}{n} f\left(i \frac{a}{n}\right)$$

is continuous for all a.

So, we can rewrite our limit:

$$\lim_{a \to x} \left(\lim_{n \to \infty} \sum_{i=1}^{n} \frac{a}{n} f\left(i \frac{a}{n}\right) \right)$$
$$= \lim_{n \to \infty} \sum_{i=1}^{n} \frac{x}{n} f\left(i \frac{x}{n}\right)$$

Summarizing with integral notation:

$$\lim_{a \to x} \int_0^a f(t) \, \mathrm{d}t = \lim_{a \to x} \left(\lim_{n \to \infty} \sum_{i=1}^{n} \frac{a}{n} f\left(i \frac{a}{n}\right) \right) = \lim_{n \to \infty} \sum_{i=1}^{n} \frac{x}{n} f\left(i \frac{x}{n}\right) = \int_0^x f(t) \, \mathrm{d}t$$

So:

$$\lim_{a \to x} \int_0^a f(t) \, \mathrm{d}t = \int_0^x f(t) \, \mathrm{d}t$$
$$\lim_{a \to x} F(a) = F(x)$$

Therefore $F(x)$ is continuous for all x.

(b) The function $F(x)$ is not twice differentiable.

False

We know:

$$F(x) = \int_0^x f(t) \, \mathrm{d}t$$

Taking the derivative of both sides:

$$\frac{\mathrm{d}}{\mathrm{d}x} F(x) = \frac{\mathrm{d}}{\mathrm{d}x} \int_0^x f(t) \, \mathrm{d}t$$

By the Fundamental Theorem of Calculus:

$$\frac{\mathrm{d}}{\mathrm{d}x} F(x) = f(x)$$

Taking a second derivative:

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} F(x) = \frac{\mathrm{d}}{\mathrm{d}x} f(x)$$

By definition, f is differentiable for all x, hence $\frac{\mathrm{d}}{\mathrm{d}x} f(x)$ exists for all x, therefore $\frac{\mathrm{d}^2}{\mathrm{d}x^2} F(x)$ exists for all x.

Therefore $F(x)$ is twice differentiable.

(c)
> The value x = 1 is a critical point of F.

True

By definition, for $x = 1$ to be a critical point of F:

$$\frac{\mathrm{d}}{\mathrm{d}x} F(1) = 0$$

From part b we know:

$$\frac{\mathrm{d}}{\mathrm{d}x} F(x) = f(x)$$

And by definition we know:

$$f(1) = 0.$$

Hence:

$$\frac{\mathrm{d}}{\mathrm{d}x} F(1) = f(1) = 0$$

Therefore, $F(x)$ has a critical point at $x = 1$.

(d)
> The function $F(x)$ takes on a local minimum at $x = 1$.

True

From part b we know:

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} F(x) = \frac{\mathrm{d}}{\mathrm{d}x} f(x)$$

And by definition, $f'(x) > 0$ for all x. Hence:

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} F(x) = f'(x) > 0$$
$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} F(x) > 0$$

Therefore, $F(x)$ is concave up for all x.

From part c we know $F(x)$ has a critical point at $x = 1$.

By the second derivative test, if a function has a critical point at $x = c$ and is concave up for all x including $x = c$, it must therefore have have a local minima at $x = c$.

Therefore $F(x)$ has a local minimum at $x = 1$.

(e)
> $F(1) > 0$.

False

Proof by counterexample.

Let $f(x) = x - 1$. This satisfies our definition of f. f is differentiable.

$$f(1) = 1 - 1 = 0$$
$$f'(x) = 1 > 0$$

Setting:

$$F(x) = \int_0^x f(t)\,\mathrm{d}t = \int_0^x t - 1\,\mathrm{d}t$$

We can solve:

$$F(x) = \frac{1}{2}x^2 - x$$

$$F(1) = \frac{1}{2}(1)^2 - (1) = -0.5$$

Therefore there exists at least one example of some $f(x)$ for which $F(1) < 0$, so the statement $F(1) > 0$ is false.

Another proof is included for fun below.

Since by definition:

$$F(x) = \int_0^x f(t)\,\mathrm{d}t$$

$F(x)$ represents the signed area under the curve of $f(t)$ from $t = 0$ to $t = x$.

We know that $f'(t) > 0$ for all t, therefore $f(t)$ is always increasing as t approaches ∞.

We also know, $f(1) = 0$.

If a function is always increasing as t approaches ∞, it must be always decreasing as t approaches $-\infty$. As a result for all $x < t$:

$$f(x) < f(t)$$

Therefore, for all $x < 1$,

$$f(x) < f(1) = 0$$
$$f(x) < 0$$

Since the area under a curve below 0 is negative, and all sections of $f(x)$ are negative for all $x < 1$ (and at $x = 1$, $f(x) = 0$), it is intuitive that the area under the curve of $f(t)$ from $t = 0$ to $t = 1$ should be negative.

Therefore:

$$0 > \int_0^1 f(t)\,\mathrm{d}t = F(1)$$
$$\therefore F(1) < 0$$

> Based on your answers above, make a rough sketch of the graph of $F(x)$.

As in part c, let $f(x) = x - 1$. This satisfies our definition. f is differentiable.

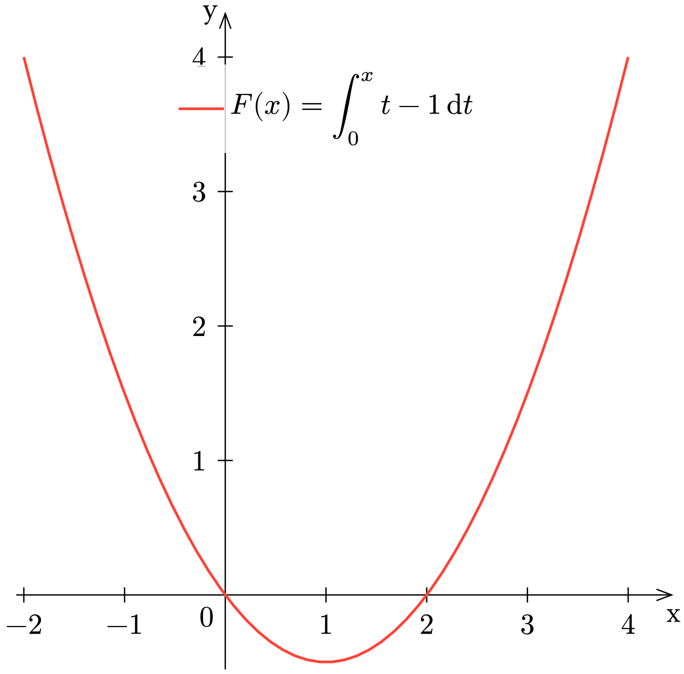
$$f(1) = 1 - 1 = 0$$
$$f'(x) = 1 > 0$$

Setting:

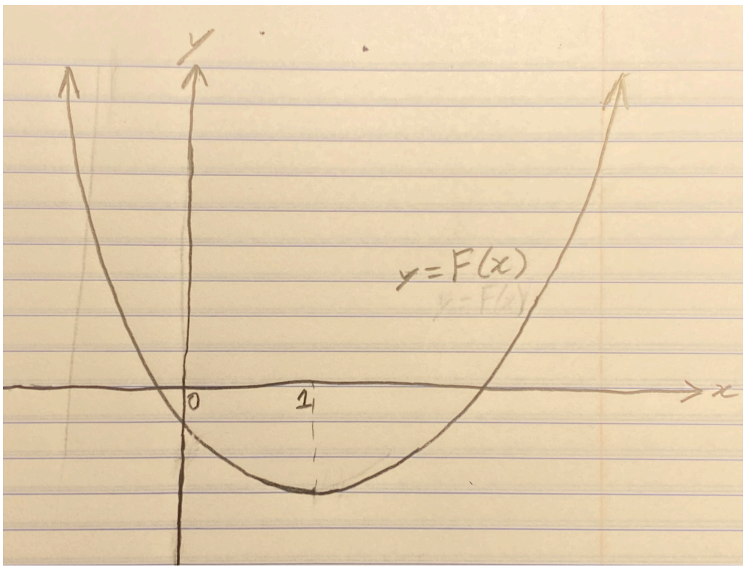
$$F(x) = \int_0^x f(t)\,\mathrm{d}t = \int_0^x t - 1\,\mathrm{d}t$$

We can solve:

$$F(x) = \frac{1}{2}x^2 - x$$



For completeness, a hand-drawn sketch of a general $F(x)$ is also included:



2. Recall in class we proved that $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$. In this problem, we will construct a proof of $S = \sum_{i=1}^{k} i^2$

(a) Calculate the value of S for $k = 1, 2, 3$ and 4.

$$\begin{aligned}
S_{k=1} &= \sum_{i=1}^{1} i^2 = 1^2 &&= 1 \\
S_{k=2} &= \sum_{i=1}^{2} i^2 = 1^2 + 2^2 &&= 5 \\
S_{k=3} &= \sum_{i=1}^{3} i^2 = 1^2 + 2^2 + 3^2 &&= 14 \\
S_{k=4} &= \sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 &&= 30
\end{aligned}$$

(b) Calculate the value of $\frac{k(k+1)(2k+1)}{6}$ for $k = 1, 2, 3$, and 4

$$\begin{aligned}
k = 1 \qquad & \frac{1(1+1)(2(1)+1)}{6} = 1 \\
k = 2 \qquad & \frac{2(2+1)(2(2)+1)}{6} = 5 \\
k = 3 \qquad & \frac{3(3+1)(2(3)+1)}{6} = 14 \\
k = 4 \qquad & \frac{4(4+1)(2(4)+1)}{6} = 30
\end{aligned}$$

(c) Show that $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$ by constructing a proof by induction by:

i. showing the equality is true for the base case (i.e., $k = 1$).
ii. for each positive integer N, the equality holds for $k = N$ implies the equality for $k = N + 1$.

Be sure to end your proof with an appropriate conclusion statement (i.e., "Therefore, by induction, ...").

Lemma 1 : $\boldsymbol{\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}}$ **holds for** $\boldsymbol{k = 1}$**.**

$$\begin{array}{cc}
\text{Left Hand} & \text{Right hand} \\
\sum_{i=1}^{1} i^2 = 1^2 = 1 & \frac{1(1+1)(2(1)+1)}{6} = 1
\end{array}$$

$$\therefore \sum_{i=1}^{1} i^2 = \frac{1(1+1)(2(1)+1)}{6}$$

Therefore $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$ holds for $k = 1$.

Lemma 2 : For each positive integer N, the equality holding for $k = N$ implies the equality for $k = N + 1$.

Let $f(x) = \sum_{i=1}^{x} i^2$. Let $g(x) = \frac{x(x+1)(2x+1)}{6}$

Let n be some positive integer such that the equality $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$ holds.

Therefore $f(n) = g(n)$

Consider $f(n+1)$.

$$\begin{aligned} f(n+1) &= \sum_{i=1}^{n+1} i^2 = \left(\sum_{i=1}^{n} i^2\right) + (n+1)^2 \\ f(n+1) &= f(n) + (n+1)^2 \end{aligned}$$

Then consider $g(n+1)$.

$$\begin{aligned} g(n+1) &= \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6} \\ &= \frac{(2n^3+9n^2+13n+6)}{6} \\ &= \frac{(2n^3+3n^2+1n)+(6n^2+12n+6)}{6} \\ &= \frac{n(2n^2+3n+1)}{6} + (n^2+2n+1) \\ &= \frac{n(2n+1)(n+1)}{6} + (n+1)^2 \\ &= g(n) + (n+1)^2 \end{aligned}$$

Putting them together, by definition:

$$\begin{gathered} f(n) = g(n) \\ f(n) + (n+1)^2 = g(n) + (n+1)^2 \\ \therefore f(n+1) = g(n+1) \end{gathered}$$

Therefore for every $N \in \mathbb{Z}^+$, $f(N) = g(N)$ implies that $f(N+1) = g(N+1)$. i.e the equality holding for $k = N$ implies that it holds for $k = N + 1$

Therefore by induction, since $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$ holds for $k = 1$ and this equality holding for $k = N$ implies that it holds for $k = N + 1$ provided that $N \in \mathbb{Z}^+$:

$$S = \sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$$

holds for all $k \in \mathbb{Z}^+$

3. Let $f(x)$ be an integrable function over the interval $[a, b]$. Choose an integer N and let

$$x_k = a + k\Delta x, \quad k = 0, 1, 2, ..., N \text{ where } \Delta x = \frac{b-a}{N}$$

The trapezoid rule is an approximation of the definite integral:

$$\int_a^b f(x)\, \mathrm{d}x \approx \Delta x \sum_{k=1}^{N} \frac{f(x_k) + f(x_{k-1})}{2},$$

where $\Delta x = \frac{b-a}{N}$

A bound on the error is given by the error formula

$$\left| \int_a^b f(x)\, \mathrm{d}x - \frac{b-a}{N} \sum_{k=1}^{N} \frac{f(x_k) + f(x_{k-1})}{2} \right| \leq \frac{(b-a)^3}{12N^2} K_2$$

where K_2 is any number such that $|f''(x)| \leq K_2$ for all $x \in [a, b]$.

Consider the Fresnel integral

$$\int_0^{\sqrt{\frac{\pi}{2}}} \sin(x^2)\, \mathrm{d}x$$

(a) Write out and calculate the Left and Right Riemann sums for $N = 4$.

Left Riemann Sum

$$\int_0^{\sqrt{\frac{\pi}{2}}} \sin(x^2)\, \mathrm{d}x \approx \sum_{k=1}^{N} \sin(((k-1)\Delta x)^2)\Delta x$$

$$= \sum_{k=1}^{4} \sin\left(\left((k-1)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right)\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) =$$

$$= \sin\left(\left(\frac{0\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) + \sin\left(\left(\frac{1\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) + \sin\left(\left(\frac{2\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) + \sin\left(\left(\frac{3\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right)$$

$$= 0.3928234382665112$$

Right Riemann Sum

$$\int_0^{\sqrt{\frac{\pi}{2}}} \sin(x^2)\, \mathrm{d}x \approx \sum_{k=1}^{N} \sin(x_k^2)\Delta x$$

$$= \sum_{k=1}^{4} \sin\left(\left(k\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right)\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) =$$

$$= \sin\left(\left(\frac{1\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) + \sin\left(\left(\frac{2\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) + \sin\left(\left(\frac{3\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) + \sin\left(\left(\frac{4\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right)$$

$$= 0.7061519725953862$$

(b) Write our and calculate the Trapezoid rule for $N = 4$.

$$\int_a^b \sin(x^2)\,\mathrm{d}x \approx \Delta x \sum_{k=1}^{N} \frac{\sin(x_k^2) + \sin(x_{k-1}^2)}{2}$$

$$= \left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right) \sum_{k=1}^{4} \frac{\sin\left(\left(k\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right)\right)^2\right) + \sin\left(\left((k-1)\left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right)\right)^2\right)}{2}$$

$$= \left(\frac{\sqrt{\frac{\pi}{2}}}{4}\right)\left(\frac{\sin\left(\left(\frac{1\sqrt{\frac{\pi}{2}}}{4}\right)^2\right) + \sin\left(\left(\frac{(1-1)\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)}{2} + \frac{\sin\left(\left(\frac{2\sqrt{\frac{\pi}{2}}}{4}\right)^2\right) + \sin\left(\left(\frac{(2-1)\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)}{2} + \frac{\sin\left(\left(\frac{3\sqrt{\frac{\pi}{2}}}{4}\right)^2\right) + \sin\left(\left(\frac{(3-1)\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)}{2} + \frac{\sin\left(\left(\frac{4\sqrt{\frac{\pi}{2}}}{4}\right)^2\right) + \sin\left(\left(\frac{(4-1)\sqrt{\frac{\pi}{2}}}{4}\right)^2\right)}{2}\right)$$

$$= 0.5494877054309487$$

(c) Find an appropriate value of K_2 for the Fresnel integral. Be sure to justify your answer.

By definition "K_2 is any number such that $|f''(x)| \leq K_2$ for all $x \in [a, b]$."

In our case, $f(x) = \sin(x^2)$ and $a = 0, b = \sqrt{\frac{\pi}{2}}$. Therefore, we must bound $f''(x)$ on $\left[0, \sqrt{\frac{\pi}{2}}\right]$:

$$f''(x) = \frac{\mathrm{d}^2}{\mathrm{d}x^2}\sin(x^2) = \frac{\mathrm{d}}{\mathrm{d}x}2x\cos(x^2) = 2\cos(x^2) - 4x^2\sin(x^2)$$

Let $g(x) = f''(x) = 2\cos(x^2) - 4x^2\sin(x^2)$

In other words, we must find $\max(|g(x)|)$ on $\left[0, \sqrt{\frac{\pi}{2}}\right]$.

First, we can find critical points:

$$\begin{aligned}
0 &= \frac{\mathrm{d}}{\mathrm{d}x}g(x) \\
&= \frac{\mathrm{d}}{\mathrm{d}x}(2\cos(x^2) - 4x^2\sin(x^2)) \\
&= -4x\sin(x^2) - 8x\sin(x^2) - 8x^3\cos(x^2) \\
&= -12x\sin(x^2) - 8x^3\cos(x^2) \\
0 &= -4x(3\sin(x^2) + 2x^2\cos(x^2)) \\
0 &= 3\sin(x^2) + 2x^2\cos(x^2) \qquad x = 0
\end{aligned}$$

We now must show that $g(x)$ has no more critical points other than $x = 0$. In other words, showing that $3\sin(x^2) + 2x^2\cos(x^2) \neq 0$ for all $x \in \left(0, \sqrt{\frac{\pi}{2}}\right]$.

At $x = \sqrt{\frac{\pi}{2}}$, $3\sin(x^2) + 2x^2\cos(x^2) = 3$. So now we only have to check $x \in \left(0, \sqrt{\frac{\pi}{2}}\right)$

For all $x \in \left(0, \sqrt{\frac{\pi}{2}}\right)$:

$$x^2 > 0$$
$$\sin(x^2) > 0$$
$$\cos(x^2) > 0$$

Hence:

$$3\sin(x^2) > 0$$
$$2x^2\cos(x^2) > 0$$
$$3\sin(x^2) + 2x^2\cos(x^2) > 0$$
$$3\sin(x^2) + 2x^2\cos(x^2) \neq 0$$

Therefore there are no solutions to:

$$0 = 3\sin(x^2) + 2x^2\cos(x^2) \text{ for } x \in \left(0, \sqrt{\frac{\pi}{2}}\right)$$

So $g(x)$ has only one critical point at $x = 0$, and two end points at $x = 0, \sqrt{\frac{\pi}{2}}$ on $\left[0, \sqrt{\frac{\pi}{2}}\right]$.

$$g(0) = 2\cos(0^2) - 4(0^2)\sin(0^2) = 2$$
$$g\left(\sqrt{\frac{\pi}{2}}\right) = 2\cos\left(\sqrt{\frac{\pi}{2}}^2\right) - 4\left(\sqrt{\frac{\pi}{2}}^2\right)\sin\left(\sqrt{\frac{\pi}{2}}^2\right) = -2\pi$$

Therefore, the global extrema of $g(x)$ on $\left[0, \sqrt{\frac{\pi}{2}}\right]$ are $g(0) = 2$ and $g\left(\sqrt{\frac{\pi}{2}}\right) = -2\pi$. So:

$$\max(|g(x)|) = 2\pi$$
$$\max(|f''(x)|) = 2\pi$$
$$|f''(x)| \leq 2\pi$$

So a good value of K_2 would be:

$$K_2 = 2\pi$$

(d)

> Find the value of (the smallest) N which guarantees the trapezoid rule approximates the Fresnel Integral with errors less than 10^{-1} and 10^{-2}. Be sure to justify your answers.

The bound on the error from the trapezoid rule is:

$$\left|\int_a^b f(x)\,\mathrm{d}x - \frac{b-a}{N}\sum_{k=1}^{N}\frac{f(x_k) + f(x_{k-1})}{2}\right| \leq \frac{(b-a)^3}{12N^2}K_2$$

In our case, $f(x) = \sin(x^2)$ and $a = 0, b = \sqrt{\frac{\pi}{2}}$.

$$\left|\int_0^{\sqrt{\frac{\pi}{2}}} \sin(x^2)\,\mathrm{d}x - \frac{\sqrt{\frac{\pi}{2}}}{N}\sum_{k=1}^{N}\frac{\sin(x_k^2) + \sin(x_{k-1}^2)}{2}\right| \leq \frac{\left(\sqrt{\frac{\pi}{2}}\right)^3}{12N^2}K_2$$

Therefore we can just find values of N for which:

$$\frac{\left(\sqrt{\frac{\pi}{2}}\right)^3}{12N^2}K_2 < 10^{-1} \qquad \text{and} \qquad \frac{\left(\sqrt{\frac{\pi}{2}}\right)^3}{12N^2}K_2 < 10^{-2}$$

Starting off with 10^{-1}:

$$\frac{\left(\sqrt{\frac{\pi}{2}}\right)^3}{12N^2}K_2 < 10^{-1}$$

$$\frac{1}{N^2} < \frac{(12)(10^{-1})}{K_2\left(\sqrt{\frac{\pi}{2}}\right)^3} \qquad 12, \left(\sqrt{\frac{\pi}{2}}\right)^3, K_2 > 0$$

$$N^2 > \frac{K_2\left(\sqrt{\frac{\pi}{2}}\right)^3}{(12)(10^{-1})}$$

$$N > \sqrt{\frac{K_2\left(\sqrt{\frac{\pi}{2}}\right)^3}{(12)(10^{-1})}}$$

With $K_2 = 2\pi$ from part c:

$$N > \sqrt{\frac{(2\pi)\left(\sqrt{\frac{\pi}{2}}\right)^3}{(12)(10^{-1})}} \approx 3.21062230801$$

Therefore, since $N \in \mathbb{Z}$ the smallest N for which the approximation is within 10^{-1} is:

$$\boxed{N = 4}$$

Similarly, for 10^{-2}:

$$N > \sqrt{\frac{(2\pi)\left(\sqrt{\frac{\pi}{2}}\right)^3}{(12)(10^{-2})}} \approx 10.1528791998$$

Therefore the smallest N for which the approximation is within 10^{-2} is:

$$\boxed{N = 11}$$

4. > In class, we proved Part 2 of the Fundamental Theorem of Calculus, which states that if f is a continuous function on the interval $[a, b]$ and F is any antiderivative of f, then
>
> $$F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x$$
>
> Here, you will construct another proof of Part 2 of the Theorem.

(a) > Divide the interval $[a, b]$ in n subintervals with endpoints $a = x_0 < x_1 < x_2 < \ldots < x_n = b$. Show that
>
> $$F(b) - F(a) = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1}))$$

Let $a = x_0 < x_1 < x_2 < \ldots < x_n = b$

Consider the sum:

$$\sum_{i=1}^{n}(F(x_i) - F(x_{i-1}))$$

Expanding, we get:

$$\sum_{i=1}^{n}(F(x_i) - F(x_{i-1}))$$

$$= \left(\sum_{i=1}^{n} F(x_i)\right) - \sum_{i=1}^{n} F(x_{i-1}))$$

$$= (F(x_1) + F(x_2) + \dots + F(n-1) + F(x_n)) - (F(x_0) + F(x_1) + F(x_2) + \dots + F(x_n - 1))$$

All terms except for $F(x_0)$ and $F(x_n)$ cancel, leaving:

$$\sum_{i=1}^{n}(F(x_i) - F(x_{i-1})) = F(x_n) - F(x_0)$$

By definition $x_n = b$ and $x_0 = a$, therefore

$$\boxed{\sum_{i=1}^{n}(F(x_i) - F(x_{i-1})) = F(b) - F(a)}$$

(b)

> Now suppose that F is any antiderivative of f. Show that there exists a number c_i in each interval $[x_{i-1}, x_i]$ such that
>
> $$F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1})$$

Let $g(x)$ be a function continuous on the interval $[j, k]$.

Let $G(x)$ be any antiderivative of a function $g(x)$.

By definition, $G(x)$ is differentiable on $[j, k]$, and since differentiability implies continuity, is also continuous on $[j, k]$.

By the Mean Value Theorem, there must exist some $u \in (j, k)$ such that:

$$\frac{G(k) - G(j)}{k - j} = G'(u)$$

Since by definition, $G'(u) = g(u)$, we can rearrange:

$$G(k) - G(j) = g(u)(k - j)$$

Now we can show that this holds for our original functions f and F.

$f(x)$ is a function continuous on the interval $[a, b]$ and is therefore continuous on $[x_{i-1}, x_i]$ as by definition, $a \le x_{i-1} < x_i \le b$.

$F(x)$ is an antiderivative of a function $f(x)$.

Therefore, $f(x)$ and $F(x)$ on $[x_{i-1}, x_i]$ is analogous to $g(x)$ and $G(x)$ on $[j, k]$. Therefore, there must exist some c_i in (x_{i-1}, x_i) such that:

$$F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1})$$

(c) Now build an appropriate Riemann sum for f on $[a, b]$ and show that

$$F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x$$

We have shown that:

$$F(b) - F(a) = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1}))$$

And that there exists some $c_i \in (x_i, x_{i-1})$ such that

$$F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1})$$

for any $f(x)$ continuous on the interval $[a, b]$, any antiderivative of $f(x)$: $F(x)$, and any intervals $a = x_0 < x_1 < x_2 < \ldots < x_n = b$.

Combining these statements we get:

$$F(b) - F(a) = \sum_{i=1}^{n} (f(c_i)(x_i - x_{i-1}))$$

Taking the limit of both sides as n approaches ∞:

$$\lim_{n\to\infty} (F(b) - F(a)) = \lim_{n\to\infty} \left(\sum_{i=1}^{n} (f(c_i)(x_i - x_{i-1})) \right)$$

$$F(b) - F(a) = \lim_{n\to\infty} \left(\sum_{i=1}^{n} (f(c_i)(x_i - x_{i-1})) \right)$$

Noting that the right hand is equivalent to the limit of the Riemann sum of $f(x)$ as the number of intervals approaches infinity, we can rewrite it in integral notation, since for any function $g(x)$ defined on any interval $[u, v]$:

$$\lim_{n\to\infty} \sum_{i=1}^{n} g(x + i\Delta x)\Delta x = \int_u^v g(x)$$

Where $\Delta x = \dfrac{u - v}{n}$.

In our case $\Delta x = x_i - x_{i-1}$. By definition c_i exists in the interval (x_i, x_{i-1}), therefore we can substitute $x + i\Delta x$ as c_i as this equality holds as long as the sampled point of the function lies within each interval of the sum (i.e the limit of left, right, and midpoint Riemann sums are equivalent).

So we can rewrite:

$$\lim_{n\to\infty} \left(\sum_{i=1}^{n} (f(c_i)(x_i - x_{i-1})) \right) = \int_{x_0}^{x_n} f(x) = \int_a^b f(x)$$

Therefore:

$$\boxed{F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x}$$